

Kianna Ng

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Research Interests

Computer Vision, Robotic Perception, Human-Centered AI, Multitask Representation Learning, Autonomous Driving Safety, Multimodal Sensing

Education

University of California, Merced Aug 2025 – Present Ph.D. Electrical Engineering and Computer Science
Advisor: Dr. Ross Greer
Research Focus: Computer Vision, Robotic Perception, Human-Centered AI
University of Southern Denmark Aug 4 – 22, 2026 Elite Summer School in Robotics and Entrepreneurship
Odense, Denmark
University of California, Merced Aug 2020 – May 2024 B.S. Computer Science & Engineering GPA: 3.7

Publications

Investigating the Effects of Self-selected Pleasant Scents on Text Composition and Transcription Performance
W. Haw, Y. Ren, K. Ng, A. Arif — ACM CHI 2024
doi.org/10.1145/3613904.3642445

Experience

Graduate Student Researcher — Mi3 Lab, UC Merced 2025 – Present

- Research autonomous driving safety, robotic perception, and human-machine communication
- Develop computer vision systems for work-zone detection and dynamic speed limit recognition
- Design multitask deep learning models for driver-state estimation using gaze and emotion
- Training large-scale deep learning models on GPU clusters using Slurm and PyTorch

Undergraduate Researcher — IoT4Ag Robotics Lab, UC Merced 2023 – 2024

- Trained TensorFlow classifier on infrared imagery for robotic irrigation control
- Integrated LoRa communication with ROS2 and embedded systems
- Assisted testing navigation stacks on Husky UGV with GPS localization

Backend Developer — LLM Operations Tools 2024

- Built Slack-integrated chatbot using Python, Flask, and OpenAI APIs
- Won 1st Place — UC Merced Senior Capstone Industry Competition

Awards & Honors

1st Place — ESV International Student Competition 2026
Enhanced Safety of Vehicles (ESV) International Technical Conference
1st Place — UC Merced Senior Capstone Industry Competition 2024

Projects

D3C — Distributed Driving Data Collective

- Designing distributed smartphone sensing platform for multimodal driving data collection
- Implementing real-time synchronization of video, IMU, GPS, and audio streams
- Building WebSocket-based ingestion pipeline for large-scale driver behavior datasets

Teaching

Teaching Assistant — UC Merced 2025 – Present
Courses: Algorithm Design and Analysis, Discrete Mathematics

- Led weekly discussions, held office hours, and graded programming assignments

Technical Skills

Languages: Python, C++ , Java, JavaScript **Frameworks/Tools:** ROS2, TensorFlow, Flask, Git, Linux

Activities

Speaker / Outreach Volunteer — Society of Women Engineers (SWE)
Historian / Photographer — Hmong Student Association